Diego Strebel

Exercise Sheet 4

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Give an example of a non-empty set A and a relation σ on A satisfying the following properties, or prove why it cannot exist.

1. Reflexive and irreflexive.
2. Symmetric and antisymmetric.
3. Symmetric, transitive, and reflexive.
4. Asymmetric and symmetric.
5. Reflexive, symmetric, and antisymmetric.
6. Irreflexive and symmetric.
7. Neither reflexive nor irreflexive.

Exercise 2.
Let $A = \{1, 2, 3, 4, 5, 6\}$. Write out the relations corresponding to $|$, $\nmid$, $>$, and $\leq$, and express them in graph representation.

Exercise 3.
Prove or disprove: If σ and θ are relations on a set A, then

$$(\sigma \cup \theta)^* = \sigma^* \cup \theta^*.$$

Exercise 4.
Prove or disprove: If σ is a partial order on a set A, then its inverse relation σ^{-1} is also a partial order on A.

Exercise 5.
Let

$$P = \{\{0\}, \{-1, 1\}, \{-2, 2\}, \{-3, 3\}, \ldots\}$$

be a partition of $\mathbb{Z}$. Define an equivalence relation θ such that $\mathbb{Z}/\theta = P$.

Exercise 6.
Let σ be a relation on $\mathbb{Z}$ defined by

$$x\sigma y \iff x^3 + y^3 \text{ is even.}$$

Prove that σ is an equivalence relation and describe its equivalence classes.

Exercise 7.
Let σ be a relation on $\mathbb{R} \setminus \{0\}$ defined by

$$x\sigma y \iff \frac{x}{y} \in \mathbb{Q}^+.$$

Prove that σ is an equivalence relation and describe its equivalence classes. Are $\sqrt{2}$ and $\sqrt{8}$ in the same equivalence class? What about $\sqrt{2}$ and $\sqrt{3}$?

Exercise 8.
Let $A = \mathbb{N} \times \mathbb{N}$. Define the lexicographical relation $\preceq$ on A by

$$(a, b) \preceq (c, d) \iff (a < c) \text{ or } (a = c \text{ and } b \leq d).$$

Prove that $\preceq$ is a total order on A.

★ **Exercise 9.**
Prove or disprove: Every partition of a set A is equal to A/θ for some equivalence relation θ on A.

★ **Exercise 10.**
Let σ be any relation on a non-empty set A. Let $\theta = \sigma \cup \sigma^{-1} \cup \mathrm{Id}_A$.

1. Prove that θ^* is an equivalence relation.
2. Prove that θ^* is the smallest equivalence relation containing σ: if ρ is any equivalence relation on A such that $\sigma \subseteq \rho$, then $\theta^* \subseteq \rho$.